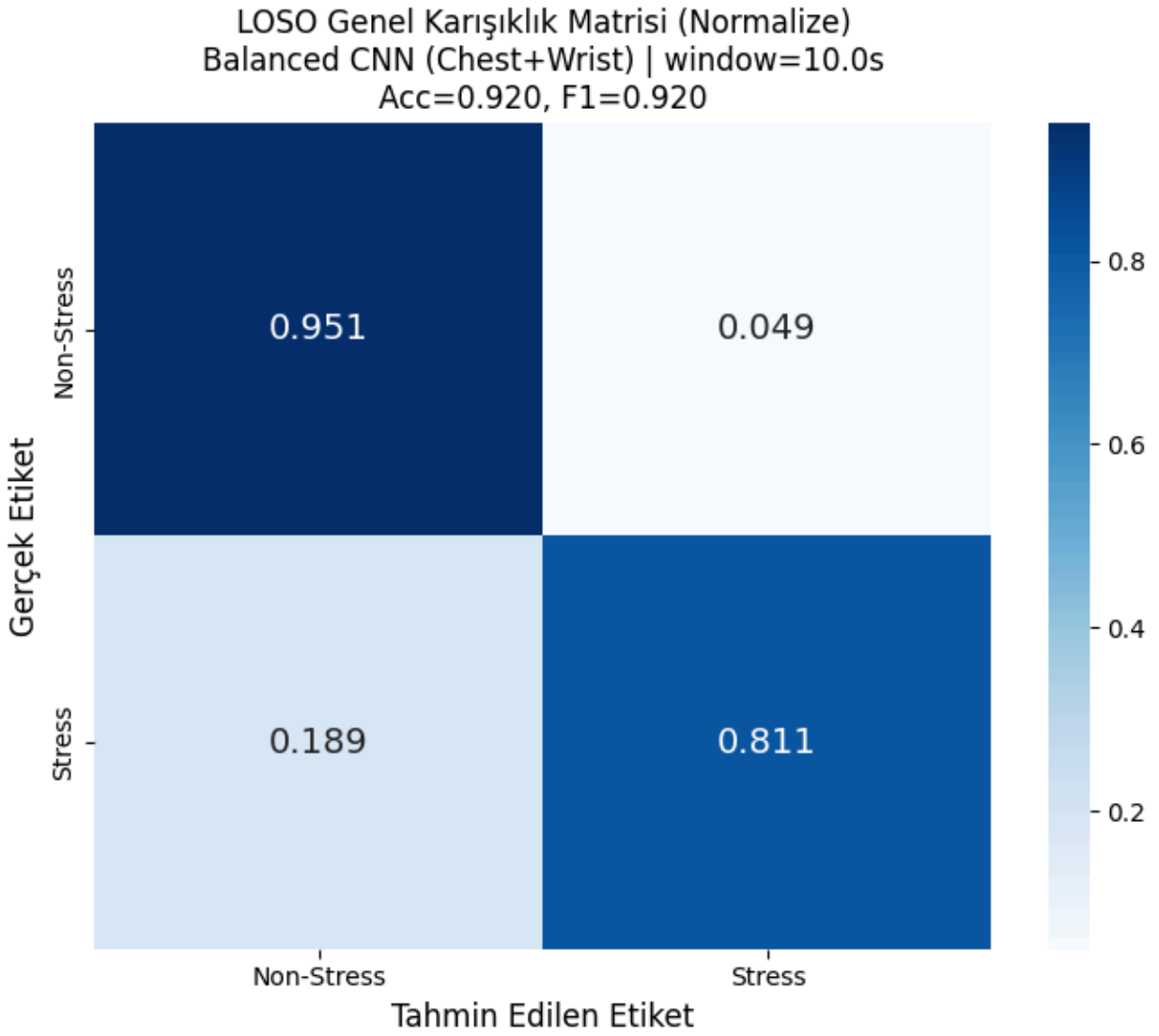
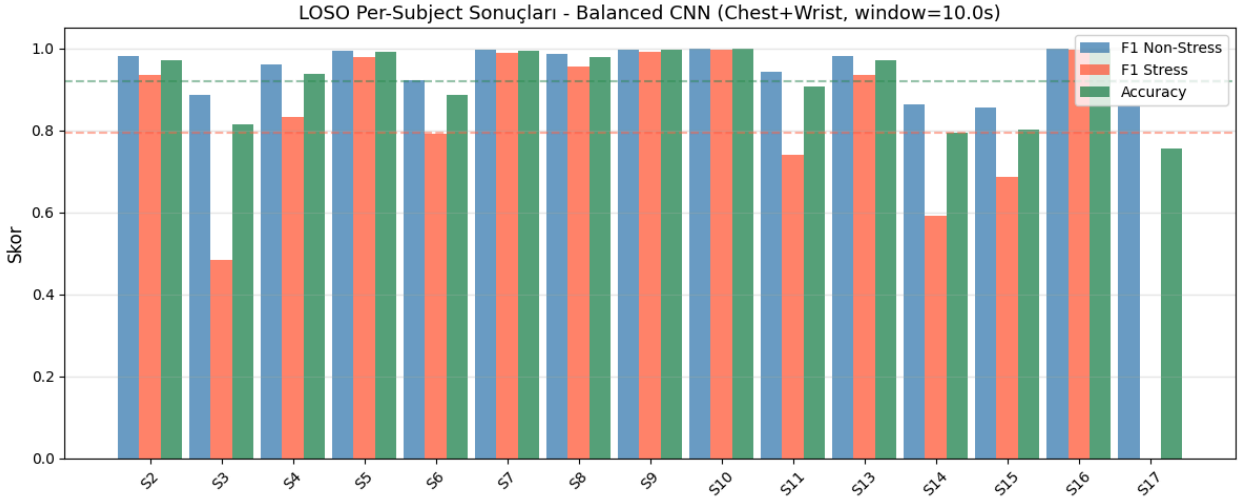
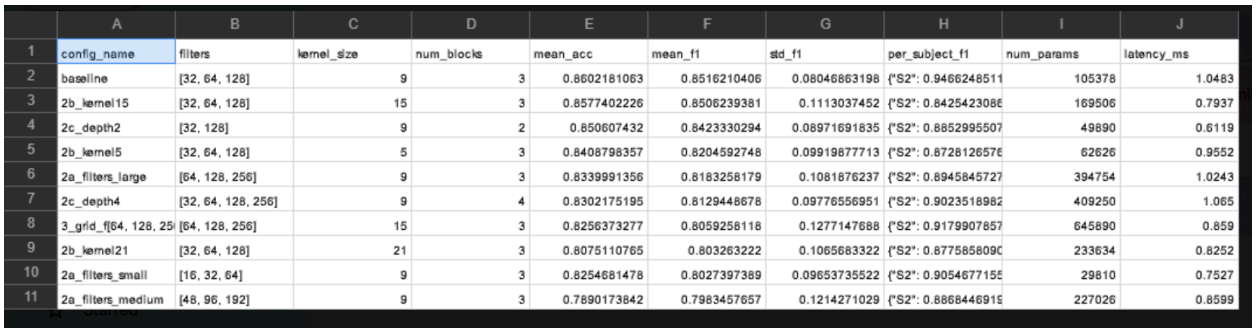
30 agustos eğitim sonuçları. (Undersampling + cross validation)



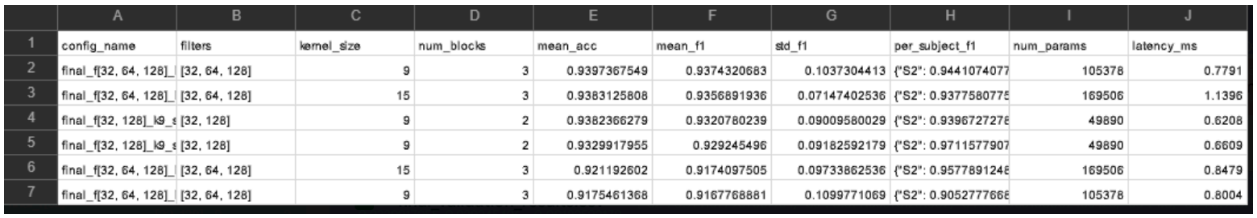


Şu anki `normalize_subject()` fonksiyonun, bir subject'in **tüm oturumunun** istatistikleriyle (`StandardScaler.fit_transform` tüm veriye) normalize ediyor. Bu, offline LOSO değerlendirmesi için sorun değil, ama **"gerçek zamanlı"** iddiasıyla çelişir — çünkü gerçek sistemde henüz gelmemiş gelecekteki verinin ortalama/std'sini bilemezsin (bu bir tür "future leakage"). Replay emülasyonunda bunu düzeltmen gerekecek, **Sabit (pre-computed) normalizasyon**: train subjectlerinden öğrenilmiş sabit mean/std'yi kullanmak, test subject'e özel hesaplama yapmamak



	A	B	C	D	E	F	G	H	I	J
1	config_name	filters	kernel_size	num_blocks	mean_acc	mean_f1	std_f1	per_subject_f1	num_params	latency_ms
2	baseline	[32, 64, 128]	9	3	0.8602181063	0.8516210406	0.08046863198	{"S2": 0.9466248511	105378	1.0483
3	2b_kernel15	[32, 64, 128]	15	3	0.8577402226	0.8506239381	0.1113037452	{"S2": 0.8425423086	169506	0.7937
4	2c_depth2	[32, 128]	9	2	0.850607432	0.8423330294	0.08971691835	{"S2": 0.8852995507	49890	0.6119
5	2b_kernel5	[32, 64, 128]	5	3	0.8408798357	0.8204592748	0.09919877713	{"S2": 0.8728126576	62626	0.9552
6	2a_filters_large	[64, 128, 256]	9	3	0.8339991356	0.8183258179	0.1081876237	{"S2": 0.8945845727	394754	1.0243
7	2c_depth4	[32, 64, 128, 256]	9	4	0.8302175195	0.8129448678	0.09776556951	{"S2": 0.9023518982	409250	1.065
8	3_grid_f[64, 128, 25	[64, 128, 256]	15	3	0.8256373277	0.8059258118	0.1277147688	{"S2": 0.9179907857	645890	0.859
9	2b_kernel21	[32, 64, 128]	21	3	0.8075110765	0.803263222	0.1065683322	{"S2": 0.8775858090	233634	0.8252
10	2a_filters_small	[16, 32, 64]	9	3	0.8254681478	0.8027397389	0.09653735522	{"S2": 0.9054677155	29810	0.7527
11	2a_filters_medium	[48, 96, 192]	9	3	0.7890173842	0.7983457657	0.1214271029	{"S2": 0.8868446919	227026	0.8599

Aday	Params	Latency	F1
depth2 [32,128] k9	~50K	~0.6ms	0.9307
baseline [32,64,128] k9	~105K	~0.89ms	0.9271
k15 [32,64,128] k15	~170K	~1.36ms	0.9265



	A	B	C	D	E	F	G	H	I	J
1	config_name	filters	kernel_size	num_blocks	mean_acc	mean_f1	std_f1	per_subject_f1	num_params	latency_ms
2	final_f[32, 64, 128]_	[32, 64, 128]	9	3	0.9397367549	0.9374320683	0.1037304413	{"S2": 0.9441074077	105378	0.7791
3	final_f[32, 64, 128]_	[32, 64, 128]	15	3	0.9383125808	0.9356891936	0.07147402536	{"S2": 0.9377580775	169506	1.1396
4	final_f[32, 128]_k9_s	[32, 128]	9	2	0.9382366279	0.9320780239	0.09009580029	{"S2": 0.9396727278	49890	0.6208
5	final_f[32, 128]_k9_s	[32, 128]	9	2	0.9329917955	0.929245496	0.09182592179	{"S2": 0.9711577907	49890	0.6609
6	final_f[32, 64, 128]_	[32, 64, 128]	15	3	0.921192602	0.9174097505	0.09733862536	{"S2": 0.9577891248	169506	0.8479
7	final_f[32, 64, 128]_	[32, 64, 128]	9	3	0.9175461368	0.9167768881	0.1099771069	{"S2": 0.9052777668	105378	0.8004

Shap And Lime Optimization

LIME — Sıra 1: `num_samples` (segments=10, kernel_width=0.25 sabit)

#	Yapılandırma	Gecikme ort/p95 (ms)	Faithfulness	Monotonicity	Sensitivity	Kalite	Hız
1	samples250	105.3 / 211.4	0.2219	0.5925	0.0274	−4.9%	2.3×
2	**samples500**	**104.3 / 162.8**	**0.2302**	**0.6025**	**0.0266**	**−1.3%**	**2.4×**
3	samples1000	245.6 / 376.3	0.2333	0.5938	0.0264	en iyi	1.0×

LIME — Sıra 2: `num_segments` (num_samples=500, kernel_width=0.25 sabit)

#	Yapılandırma	Gecikme ort/p95 (ms)	Faithfulness	Monotonicity	Sensitivity	Kalite	Hız
1	segments5	67.5 / 71.9	0.2180	0.6175	0.0217	−8.2%	2.6×
2	segments10	151.1 / 268.1	0.2302	0.6025	0.0266	−3.0%	1.1×
3	**segments20**	**172.9 / 353.4**	**0.2375**	**0.5862**	**0.0357**	**en iyi**	**1.0×**

LIME — Sıra 3: `kernel_width` (num_samples=500, num_segments=20 sabit)

#	Yapılandırma	Gecikme ort/p95 (ms)	Faithfulness	Monotonicity	Sensitivity	Kalite	Hız
1	kw_x0.5	104.8 / 159.4	0.1887	0.6700	0.0147	−21.0%	1.7×
2	kw_default	181.1 / 358.0	0.2375	0.5862	0.0357	−0.6%	1.0×
3	**kw_x2**	**152.8 / 344.5**	**0.2390**	**0.5813**	**0.0365**	**en iyi**	**1.2×**

SHAP

SHAP — Sıra 1: `background_size` (nsamples=200, method=random sabit)

#	Yapılandırma	Gecikme ort/p95 (ms)	Faithfulness	Monotonicity	Sensitivity	Kalite	Hız
1	bg10	192.3 / 232.7	0.2038	0.6262	0.1119	−9.1%	1.1×
2	bg50	207.1 / 247.7	0.2212	0.6150	0.1003	−1.4%	1.0×
3	**bg100**	**204.4 / 252.8**	**0.2243**	**0.5938**	**0.1011**	**en iyi**	**1.0×**

SHAP — Sıra 2: `nsamples` (background_size=100, method=random sabit)

#	Yapılandırma	Gecikme ort/p95 (ms)	Faithfulness	Monotonicity	Sensitivity	Kalite	Hız
1	nsamples50	46.5 / 54.9	0.2214	0.5950	0.1083	−1.4%	8.6×
2	**nsamples200**	**204.9 / 249.4**	**0.2243**	**0.5938**	**0.1011**	**−0.1%**	**2.0×**
3	nsamples400	402.2 / 487.7	0.2246	0.6025	0.1032	en iyi	1.0×

Seçiminiz: #2 (nsamples=200) — dikkat: bu sefer dengeli öneri **#1 (nsamples50)** idi, siz ondan sapıp daha pahalı olan #2'yi seçtiniz. Önceki (offline) koşuda tam tersini yapıp en ucuzu (#1) seçmiştiniz.

SHAP — Sıra 3: `background_method` (background_size=100, nsamples=200 sabit)

#	Yapılandırma	Gecikme ort/p95 (ms)	Faithfulness	Monotonicity	Sensitivity	Kalite	H
1	method_random	198.4 / 234.7	0.2243	0.5938	0.1011	−2.6%	1.(
2	**method_kmeans**	**199.3 / 240.5**	**0.2302**	**0.5625**	**0.1184**	**en iyi**	**1.(**

SONUCLAR

Son karşılaştırma (causal-normalize, nihai)

	LIME	SHAP	Fark
latency mean	152.8 ms	199.3 ms	LIME %23 daha hızlı (ortalamada)
latency p95	**344.5 ms**	**240.5 ms**	SHAP kuyrukta daha kararlı
faithfulness	**0.2390**	0.2302	LIME %3.8 daha iyi
monotonicity	**0.5813**	0.5625	LIME %3.3 daha iyi
sensitivity	**0.0365**	0.1184	**LIME 3.2× daha kararlı**